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High-Level Explanation (What Are We Building?)

You're building a **smart travel assistant**. It:

- Helps users plan trips
- Gives flight & hotel suggestions
- Checks if their budget is realistic
- Adapts recommendations to user preferences
- Has a user-friendly **web interface using Streamlit**

Tech Stack Overview

Tool	Purpose
Streamlit	Frontend (UI) for the user to interact with the travel agent
Python + OpenAI Agents SDK	Backend logic — smart decision-making
AsyncOpenAI	Talk to an OpenAI model with function calling

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| **Pydantic** | Define structured responses (like `TravelPlan`, `FlightRecommendation`, etc.) |
| **dotenv** | Load API credentials securely |

🌐 What is Streamlit?

Streamlit is a Python library to create **interactive web apps** for **data science, ML, and AI — without needing HTML, CSS, or JS**.

Think of it like:
🧑‍💻 Python Script → **Streamlit Magic** → 🌐 Web App

Key Concepts:

Concept	What It Does
`st.title`, `st.text_input`, `st.button`, etc.	Creates UI elements
`st.session_state`	Keeps data between user actions (like chat history)
`st.chat_input`	Special input for chatbot messages
`st.rerun()`	Reruns the app script after interaction
`st.markdown()`	Allows HTML/CSS for styling

🧩 How Does the Travel Agent Connect to Streamlit?

Let's break it down:

✅ 1. User Enters a Message

user_input = st.chat_input("Ask about travel plans...")